

Privacy Coins Under Viewing Key Compromise

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Abstract. Anonymity guarantees of privacy-oriented cryptocurrencies are garnering negative attention from lawmakers who view them as anti-nomic to accountability. Having recognized their potential for innovation, however, regulators may not want to outright ban privacy coins but instead seek a middle ground where financial oversight is effective, and still some privacy is maintained. Mature designs, such as Zcash, Monero, or Firo, are expected to facilitate this through so-called *viewing keys* that can be disclosed to third parties for the purpose of supervision. This paper studies which privacy guarantees continue to hold once they have been. In doing so, it fills the gap in provable anonymity guarantees for Zcash and Firo under the compromise of the incoming viewing key, while, at the same time, exposing problems with Monero. Finally, the paper shows that malicious parties may happily surrender *all* their viewing keys in each of Zcash, Monero, and Firo and yet still find very efficient ways of evading financial monitoring.

Keywords: Privacy coins · Compliance · Anamorphic cryptography

1 Introduction

Privacy enhancement is a delicate balancing act with exercising a fundamental human right on the one hand and, on the other, equipping malicious actors with powerful tools for evading justice [15]. With technology capable of granting anonymity indiscriminately to honest users and criminals alike, the increased attention legislators have been paying cryptography in recent years is justified. Of most concern, naturally, are technologies involving financial instruments such as cryptocurrencies as a domain where accountability and transparency are most desired by state actors, *e.g.*, a revenue service, and, at the same time, incentives to be dishonest are the most tangible. This concern is reflected in, *e.g.*, recent European Union legislation [70, 71] as well as various other rulings around the world [61]. As Bitcoin is now well understood to not provide any meaningful anonymity [7, 51, 64, 67], one sees a transition to more technologically sophisticated cryptocurrencies, so-called “privacy coins,” that offer stronger guarantees. This can be observed both among regular users¹ [42] and criminals [20, 21, 36, 66].

¹ For example, Zcash saw a nearly twentyfold increase in price between year 2025 and 2026. See <https://coinmarketcap.com> (accessed: 30 August 2026).

With transaction histories cryptographically hidden, blockchain intelligence services and coin blacklisting [3, 5, 6, 53, 56], both standard tools for introducing accountability to cryptocurrencies, become ineffective against privacy coins. So do KYC practices at the crypto-fiat boundary [55]. In light of this, legislators may either rush to prohibition (see, *e.g.*, [71, Article 76(3)] or [72, Article 79(1)]), risking driving usage off regulated rails and worsening visibility, or recognize their potential for good (*e.g.*, protection of individual privacy, shielding businesses from corporate espionage, *etc.*) and search for new avenues of financial oversight in the technical landscape of privacy coins.

Mature designs, such as Monero [46, 65] or Zcash [11, 40],² take note of the compliance obligations and incorporate mechanisms that facilitate some degree of oversight through *viewing keys*, trapdoors that can be disclosed to a supervising authority to grant them varying levels of insight into the subject’s activity. They can, *e.g.*, enable decryption of incoming transactions or recognizing when some asset gets spent. Little attention has been paid in the literature, however, to the privacy guarantees that are upheld in the case viewing keys are compromised. Also, despite viewing keys often being proposed to serve some vaguely defined role in financial monitoring [26, 27, 33, 58], their exact effectiveness as a compliance tool has never been evaluated. These gaps are addressed in the present paper. In particular, if viewing key disclosure is to be officially mandated, it is crucial to (1) formally show viewing keys do not “overstep their boundaries” and indeed grant only the capability they were designed to grant, nothing more, as well as (2) evaluate to what degree bad actors can nonetheless evade oversight built around them. Notably, the former is of independent interest, as viewing keys are already routinely disclosed to untrusted servers to lift from lightweight wallets the burden of scanning the blockchain for inbound transfers [75].

In addressing the latter, this work picks up on and elaborates the analogy observed by Cinal *et al.* in a recent case study of Monero [25], which drew a comparison between the setting of financial oversight of cryptocurrencies and that of *anamorphic cryptography* [48, 49, 59]. Indeed, anamorphic cryptography is a field concerned with “possible abuses, misuses, and new uses of cryptosystems beyond their primary goals” [49] in the presence of a *dictator* that may demand participants in the system give up their private keys. While, in the case of cryptocurrencies, surrendering *all* private keys would amount to granting the dictator full custody over one’s assets, a more appropriate notion, as argued in [25], is that the dictator instead demands to see only the viewing keys. This model applies most prominently to a trading platform (an exchange) exercising due diligence as prescribed by law [68, 70, 71]. Disclosure of viewing keys could be stipulated in the exchange’s terms of service under, *e.g.*, anti-money laundering (AML) law [72]. Any cryptocurrency bought at the exchange would then only be released to an external wallet once the viewing keys have been surrendered and proved consistent (*cf.* [68, 70]). At the same time, regulated entities (such

² Capitalized at, respectively, 8.84 and 14.12 billion US dollars as of this writing according to <https://coinmarketcap.com> (accessed: 30 August 2026).

as said exchange) may be obligated to submit their, say, Zcash transactions to an audit, *e.g.*, for tax reporting purposes, in which case the auditor could demand viewing keys to ensure all transfers are correctly accounted for [10, 22, 25]. Finally, the model applies to lightweight wallets surrendering viewing keys to powerful servers [75]. More analogies between financial oversight and anamorphic cryptography can be observed. First, the financial regulator is assumed to not be qualified to establish a new protocol design that would admit the desired balance of accountability and privacy (see, *e.g.*, [28, 38]) but is instead expected to endorse an existing one (*cf.* [5, 6, 10, 25, 56]). Indeed, cryptocurrencies have great inertia to them and outcompeting existing solutions may often not be feasible due to strong network effects, so it is “important to explore the practical options for taming the systems that already exist” [5]. To use the language of Chatzigiannis *et al.* [22], we shall therefore mostly (except in Section 4.2) be interested in *auditability functionalities* as opposed to *accountability policies* which would necessarily entail changes to the consensus layer (*cf.* [22, Insight 6]). Finally, anamorphic cryptography strives for plausible deniability and assumes a “rational dictator” that will not prosecute their subjects groundlessly. Likewise, democratic societies usually require that suspicion be established before conducting an investigation, lest it be deemed unlawful and any evidence collected become inadmissible in court [30]. In light of the above, the present paper will consistently use the term *dictator* even when dealing with, say, an auditor in a democratic setting. Similarly, for clarity, it will refer to their auditees as *subjects*.

1.1 Paper Contributions

The paper evaluates which privacy coin guarantees continue to hold in the case viewing keys are surrendered to a supervising authority (the dictator) and the limitations of such disclosure in enforcing accountability. To this end, it (1) formalizes in Section 3 spend indistinguishability relative to a set of viewing keys disclosed, and, in particular, (2) studies whether it holds (as it is implicitly expected to by the users) relative to standard so-called “incoming” viewing keys in existing privacy coins. Notably, in the case of Monero, the paper argues that no black-box reduction to the anonymity claim of the underlying ring signature is possible despite this being taken for granted by the community. In Section 4, the paper (3) turns to evaluating the effectiveness of viewing key disclosure as a means for financial monitoring and demonstrates that even the disclosure of the so-called “full” viewing key is ineffective against sophisticated subjects. They may minimize the data they publish on chain to only what is strictly required by the consensus protocol and rely on off-chain (or subliminal) channels otherwise. Finally, the paper (4) proposes how the issue can be remedied in a minimally intrusive fashion and while keeping data stored on chain to the absolute minimum.

1.2 Related Work

Regulation of cryptocurrency trade, as relevant to this work, is given treatment in [5, 6, 9, 53, 56]. Balancing accountability against privacy is the focus of [10, 14, 18, 22, 28, 38, 44], with [38] in particular proposing modifications to Zcash that facilitate oversight. Anonymity guarantees of Bitcoin (or lack thereof) are surveyed in [7, 51, 55, 64, 67], while the privacy of Monero is evaluated in [32, 47, 54, 73, 74, 76] and that of Zcash in [12, 45, 63, 74]. Deuber *et al.* in [30] systematize the (often implicit) assumptions underlying analyses such as those above. Anamorphic cryptography is introduced in [48, 49, 59], while covert channels in privacy-oriented cryptocurrencies are constructed and studied in [4, 13, 25].

2 Preliminaries

2.1 Notation

Let λ denote a security parameter. A function of λ is said to be *negligible* if it decays faster than the inverse of any polynomial in λ . An algorithm is said to be *efficient* if it is probabilistic and runs in time polynomial in the size of its input (which typically also means polynomial in λ). Write $\mathbb{G}$ to denote an elliptic curve group of prime order q (with $|q| = 2\lambda$) over a prime field $\mathbb{F}_p$ and G, H, K, F to denote its generators sampled independently. Write Hash_X to denote a cryptographic hash function with range X . Write $x \leftarrow \$ X$ to denote x being sampled from a probability distribution X , *e.g.*, outputs of a probabilistic algorithm. When X is a set, it is understood that x is sampled uniformly at random from X . For consistency with reference literature, the notation for different keys used in Monero, Zcash, and Firo is based on, respectively, [46], [40] and [43], and introduced, as needed, below.

2.2 Privacy Coins

Cryptocurrencies rely on a distributed append-only ledger (a blockchain) to record transactions. For concreteness, this paper restricts its attention to cryptocurrencies based on the *unspent transaction output* (UTXO) model originally established by Bitcoin. This captures all practically relevant privacy coins.³ Transactions in the UTXO model have outputs referred to as *notes* or, if still unspent, *UTXOs*. A transaction then *spends* (consumes) one or more existing (heretofore unspent) notes as well as *mints* (creates) one or more new notes (UTXOs). Notes are associated with amounts, and a transaction is valid (hence accepted by the network and eventually appended to the ledger) if the total amount minted is less than or equal to the total amount spent (no money is created), the inputs have not been spent before, and the creator proves to have

³ The UTXO model is overall more accommodating of privacy-enhancing technologies [8, 17]. If need be, however, the spend indistinguishability notion from Section 3 can be easily modified to apply to the account-based setting as well.

been the recipient of the input notes. Normally, in, say, Bitcoin, the transaction ledger is fully transparent with transactions stored in plaintext. This is to the detriment of the privacy of the transacting parties [7, 29, 51, 64, 67]. To address this and enhance the anonymity and confidentiality of the users, privacy coins employ various cryptographic techniques.

In a privacy coin, details of a note, such as its amount or recipient’s address, are hidden using cryptographic commitments. The consistency of a transaction is then proved in zero knowledge. This includes showing that the input and output amounts balance and that the originator has spend authority over the input being spent. The links between mints and spends are also hidden using various techniques such as ring signatures [37, 43, 46, 65] or zero-knowledge proofs of membership in a Merkle tree [11, 40]. To prevent double-spending in this setting, *i.e.*, referencing the same output twice, notes have *tags* (called *key images* in Monero [46] and *nullifiers* in Zcash [40]) that must be revealed upon spending, an idea borrowed from Chaum’s e-cash [23]. A tag is derived as a pseudorandom function of the note, and the set of tags revealed so far is maintained by the nodes of the network. If a tag is ever repeated, the offending transaction is rejected.

Proving the consistency of transactions may require some extra *witnesses* associated with each input note u being spent. These include, *e.g.*, the amount attached to u or any commitment trapdoors. These are normally communicated in band, namely, the creator of the note encrypts them under the recipient’s public key and attaches the ciphertext to the note. This has the advantage that no off-chain communication or storage is needed, and the ledger is “self-contained.” A privacy-coin note u then looks as follows (compare with Appendix A):

$$(\text{note } u) = (\text{commitments } C) + (\text{encrypted witnesses } c).$$

From now on, as shorthand, we will say “decrypt a note u ” when we really mean “decrypt the ciphertext c found in u .” Importantly for the present work, the consistency of the ciphertexts published is *not* subject to the consensus mechanism, *i.e.*, the network does not verify it in any way. Users are free to mint “malformed” notes with, say, c being completely random. They may then send any witnesses out of band to the recipient.

Normally, privacy coins have every user continually *scan* the blockchain looking for notes they can successfully decrypt. In mature constructions like Monero [46], Zcash [40], or Firo [43], this computational effort can typically be offloaded by the cryptocurrency wallet (say, a smartphone app) to a more capable machine by the use of viewing keys [75]. Indeed, instead of using just a single key for everything, privacy coins separate responsibilities so that different keys are used to, *e.g.*, decrypt the note, compute the tag, and prove spend authority. The key used to prove note ownership (spend authority) is typically called the *spending key* with all other keys being various *viewing keys*. Users may choose (or be compelled to) disclose viewing keys to a third party without granting them control over their assets. Importantly, besides facilitating lightweight clients, viewing keys are often proposed by the community as a means of financial monitoring [26, 27, 33, 58], a proposal the present work sets out to evaluate.

The expectation is that the dictator, in control of their subjects’ viewing keys, should get extra insight into the subjects’ activities and be able to at least recognize incoming notes. They could then realistically demand the subjects reveal the details of these notes. For this purpose, all three of Monero, Zcash, and Firo define *incoming viewing keys* (denoted, respectively, k^v , ivk , and addr_{in} in the literature), which are studied in Section 3. The viewing key hierarchy can get more nuanced, as is the case with Zcash and Firo, which define *full viewing keys* that are used in the computation of tags. Supposedly, a dictator in the possession of these should be able to account for all inbound and outgoing transactions having to do with a particular subject. What this paper shows in Section 4, however, is that even full viewing keys are only effective when dealing with “semi-honest” subjects that use some default implementation of wallet software and not rely on out-of-band (or subliminal [4,13,25]) channels. Finally, Zcash also defines an *outgoing viewing key* ovk that can (at the discretion of the transaction originator) enable its holder to decrypt the output note and learn the intended recipient. This is again not enforced by consensus [40, Section 4.20.1].

Since viewing keys play a crucial role in this work, privacy coins that do not define any, such as Quisquis [37,57] or Mimblewimble [60], shall not be of any interest.⁴ Also, for lack of space, a more detailed introduction of Monero, Zcash, and Firo is relegated to Appendix A. Comprehensive descriptions of the full protocols can be found in [46], [40], and [43], respectively.

3 Spend Indistinguishability Relative to Viewing Keys

The most practical way to impose financial oversight on privacy-oriented cryptocurrencies, in particular those already deployed, is to obligate the subjects legally (but not technically) to submit their transactions for “clearing” with the dictator. The subjects would then have to reveal and attest to the information requested by the dictator such as the recipient or the amount [10,18,25]. Obviously, bad actors will not play by the rules and instead leverage the cryptographic guarantees of a privacy coin to hide the money flow. Viewing keys are often proposed as a means for enforcing accountability even in this setting [26,27,33,58]. For example, cryptocurrency exchanges may be monitored (by, *e.g.*, a financial supervision authority) for deposits from self-hosted wallets, justifying mandating the disclosure of the incoming viewing key [25,70]. While this enables the dictator to identify incoming funds (*i.e.*, “credits” to the exchange), they intuitively *should* remain oblivious to outgoing transfers: disclosure of the incoming viewing key is not expected to violate sender anonymity. This becomes even more of an issue when one considers another common reason to disclose incoming viewing keys, namely, scanning the ledger for inbound transfers, which is a responsibility that lightweight clients normally offload to an untrusted server. Both the above concerns: (1) detecting failure to submit a transaction to an audit and (2) not

⁴ The Omniring paper [50] mentions the possibility of extending it with viewing keys, but the details are not elaborated.

overstepping the boundaries of viewing key disclosure are captured in a unified model in this section.

The literature so far has been lacking in the evaluation of privacy coins relative to different viewing keys that the dictator can compel their subjects to disclose. Zcash Improvement Proposal number 310 [41] lists informal guarantees for the dictator regarding the consistency of the view they obtain when given the subject’s full viewing key fvk . Guarantees relative to “weaker” viewing keys down the Zcash key hierarchy [40, Section 3.1] are not considered there.⁵ What is more alarming still is that there seems to be a non-trivial gap to be filled in the provable security of Monero under viewing key compromise. Indeed, in the case viewing keys are known to the dictator, sender anonymity no longer can be reduced to chosen-target linkable anonymity of the underlying ring signature scheme [39, Definition 9] (see Definition 3 in Appendix B). We reformulate the implicit belief of the community as Conjecture 1 and elaborate on the difficulty in resolving it via a black-box reduction to ring signature security. More generally, in the cases of both Monero and Firo (for which the corresponding proof has also been missing so far), viewing keys seem to be an afterthought added on top of an otherwise sound design without redoing the analysis.

3.1 Spend Indistinguishability Model

UTXOs and Owned UTXOs. For the purposes of this paper, one must distinguish between UTXOs (more generally: notes) as perceived by the network and UTXOs as perceived by their owners. Specifically, for each UTXO u (*i.e.*, a transaction output that has not been referenced before; see Appendix A for details), define the corresponding *owned UTXO* $\bar{u}$ which, in addition to u , contains all the private keys associated with u and witnesses necessary to spend u (compare this to the concept of an *augmented ledger* in the extended version of [11]). Fixing the witnesses in the owned UTXOs “binds” the tags to UTXOs and simplifies reasoning about spends. Indeed, to say that a transaction T spends a UTXO u is to say the *tag* s of u is revealed in T (recall Section 2.2), but the network may only be convinced of the link between s and u via a zero-knowledge *argument*, the soundness of which need not hold against unbounded provers. With witnesses fixed in the owned UTXO $\bar{u}$ corresponding to u , we may simply rely in our model on a deterministic algorithm (predicate) **Spends** that decides whether a transaction T spends a UTXO u by recomputing the tag of u and comparing with the tag revealed in T .

Cryptocurrency Model and Spend Indistinguishability. The relevant (for our purposes) functionality of a cryptocurrency can then be reduced to three algorithms. **Setup**, on input security parameter in unary, returns public parameters pp . **Coinbase** produces a state of the ledger L (*e.g.*, the *note commitment tree* in

⁵ The proposal is now also outdated, not reflecting the newest version of the Zcash protocol, Orchard (Ironwood), that is of interest to this paper; see <https://github.com/zcash/zips/issues/606> (accessed: 30 August 2026).

Zcash [40, Section 3.8]) together with a UTXO set U such that all UTXOs $u \in U$ use the same amount (not strictly necessary). Finally, **Spends** takes a transaction T together with an owned UTXO $\bar{u}$ and returns 1 if T is consistent with the state of the ledger L and spends u ; otherwise, it returns 0. Notably, **Coinbase** produces the UTXOs honestly according to the *specification* of a cryptocurrency which prescribes more than is normally verified by its *consensus protocol*. For example, where the specification requires a random string be used, a malicious user may use the output of a pseudorandom function instead. The network typically has no way of telling. This is elaborated below. Compare also the definition of **Coinbase** with the Mint algorithm in [11] or [43]. In general, in prevailing designs, transaction outputs may be sampled independently of the inputs, with zero-knowledge arguments eventually linking the two (see Appendix A). It is therefore fine for us to consider outputs alone in the experiment, detached from the transactions that would have minted them in the real world. Finally, the viewing keys demanded by the dictator are expressed as a set V of functions v which, given an owned UTXO $\bar{u}$, map it to the actual viewing keys $v(\bar{u})$ associated with the owner of u .

Definition 1. A cryptocurrency $\Pi = (\text{Setup}, \text{Coinbase}, \text{Spends})$ has indistinguishable spends relative to viewing keys V if there exists an efficient (probabilistic polynomial-time) spender algorithm $\mathcal{S}$ such that, for all efficient dictators $\mathcal{D}$ and all anonymity set sizes $2 \leq n \leq \text{poly}(\lambda)$, there exists a negligible function ε such that, for all $b \in \{0, 1\}$,

$$\Pr[\text{SpendInd}_{\Pi, \text{Coinbase}, \mathcal{S}, \mathcal{D}}^b(\lambda) = (\cdot, 1)] \geq 1 - \varepsilon(\lambda),$$

and

$$\begin{aligned} & \left| \Pr[\text{SpendInd}_{\Pi, \text{Coinbase}, \mathcal{S}, \mathcal{D}}^0(\lambda) = (1, \cdot)] \right. \\ & \quad \left. - \Pr[\text{SpendInd}_{\Pi, \text{Coinbase}, \mathcal{S}, \mathcal{D}}^1(\lambda) = (1, \cdot)] \right| \leq \varepsilon(\lambda), \end{aligned}$$

where the experiment $\text{SpendInd}_{\Pi, \mathcal{A}, \mathcal{S}, \mathcal{D}}^b$ is presented in Figure 1, and the probabilities are taken over the random coins of all algorithms involved.

Definition 1 considers the following experiment. First, n UTXOs (of equal value and addressed to random recipients) $u_1, \dots, u_n$ are minted honestly, and the dictator (adversary) is asked to select two *challenge UTXOs* u_{i_0} and u_{i_1} . One of them is then to be spent by the spender algorithm $\mathcal{S}$. Notably, $\mathcal{S}$ may do it in any way it wants to as long as the transaction complies with the consensus protocol (the bit g encodes $\mathcal{S}$'s compliance). The dictator is asked to tell which one of u_{i_0} and u_{i_1} was spent given the viewing keys for *all* outstanding UTXOs, including u_{i_0} and u_{i_1} . A cryptocurrency has *indistinguishable* spends relative to viewing keys V if no efficient dictator, given access to the viewing keys V , can tell whether u_{i_0} or u_{i_1} is spent with non-negligible advantage over guessing. In that case, the dictator in particular cannot tell whether some UTXO of a given subject, call her Alice, is being spent in T or not. Such *spend attribution* is,

$\text{SpendInd}_{\Pi, \mathcal{A}, \mathcal{S}, \mathcal{D}}^b(\lambda)$	
1 :	$\text{pp} \leftarrow \text{Setup}(1^\lambda)$
2 :	$L, \bar{u}_1, \dots, \bar{u}_n \leftarrow \mathcal{A}(1^\lambda, \text{pp}, n)$
3 :	$\mathbf{v} \leftarrow \{(v(\bar{u}_1), \dots, v(\bar{u}_n)) \mid v \in V\}$
4 :	$i_0, i_1, \text{st} \leftarrow \mathcal{D}(1^\lambda, \text{pp}, L, \mathbf{v}, u_1, \dots, u_n)$
5 :	$T \leftarrow \mathcal{S}(1^\lambda, \text{pp}, L, i_0, i_1, \bar{u}_{i_b}, u_1, \dots, u_n)$
6 :	$g \leftarrow (\text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_b}) = 1 \wedge \text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_1-b}) = 0)$
7 :	$b' \leftarrow \mathcal{D}(1^\lambda, \text{st}, T)$
8 :	return (b', g)

Fig. 1. Spend indistinguishability experiment

however, a minimal requirement for any auditing functionality, as the dictator must be able to first identify any non-compliance in order to prosecute it (*cf.* [22, Insight 9]).

Remarks About the Model. Some comments about the model are in order. Note the quantifiers first: we require that there be *at least one* way for the subjects to transact (“there exists a spender algorithm $\mathcal{S}$ ”) such that, no matter what the dictator does (“for all dictators $\mathcal{D}$ ”), they cannot tell apart their subjects’ spends. Second, note that Definition 1 assumes the UTXOs $u_1, \dots, u_n$ are minted honestly via *Coinbase*. Letting the dictator $\mathcal{D}$ mint them instead, *i.e.*, setting $\mathcal{A} = \mathcal{D}$ (which would be more in line with the requirements of [41] and the ideas from [11] to some extent), would unnecessarily complicate the model, and it is not relevant for our purposes. Indeed, if being the sender of a transaction that mints u gives the dictator any advantage, the owner of u suspicious of this could “recycle” u by sending it back to themselves (*e.g.*, as part of making change in a legal transaction, *i.e.*, one cleared by the dictator). Conversely, we could allow $\mathcal{S}$ to produce the UTXOs, thus capturing more sophisticated users’ sending the witnesses for u_{i_0} and u_{i_1} completely out of band and only encrypting dummy data on chain. This weaker⁶ model is considered in Section 4. Finally, one could also fix $\mathcal{S}$ to be the canonical transaction creation algorithm complying fully with the specification (which is almost what the proofs in Appendix C do anyway), to get an even stronger model, where spends remain indistinguishable without manipulating the transaction creation process. The results would then correspond to honest users’ transacting with some default (reference) wallet implementations after their viewing keys have been compromised (*e.g.*, leaked by the server with which they were entrusted for the purpose of scanning the ledger [74]).

⁶ It makes the task of $\mathcal{D}$ much harder, thus making security (privacy) easier to achieve.

3.2 Spend Indistinguishability Today

With viewing keys often disclosed to semi-trusted servers, who may then fall prey to cyberattacks, it is crucial to evaluate the impact of any breaches on the privacy of users. Clearly, *full viewing keys* defined by Zcash and Firo (recall Section 2.2) do not afford spend indistinguishability. Indeed, their very purpose is the computation of tags, which are unambiguously associated to notes in order to prevent double-spending. (This is an example of the concordance between accountability and the consensus protocol observed in [22, 25].) More commonly, however, only incoming viewing keys are disclosed. This then warrants the evaluation of spend indistinguishability relative to them.

Theorem 1. *Zcash has indistinguishable spends in the random oracle model relative to the incoming and outgoing viewing keys $\{\text{ivk}, \text{ovk}\}$.*

The above theorem roughly corresponds to the security requirement of [40, Section 4.16]. Its proof in Appendix C.1 seems to be the first one that is publicly available, but the reference implementation’s documentation asserts spend indistinguishability (“unlinkability” therein) based on the pseudorandomness of the tag derivation function.⁷ An analogous result holds for Firo and is proved in Appendix C.2. Notably, the statement is strictly stronger than the natural corollary of ledger indistinguishability (LIND), whose proof is sketched in [43], as the dictator in the present work gets to see the incoming viewing keys of all outstanding UTXOs.

Theorem 2. *Firo has indistinguishable spends in the random oracle model relative to the incoming viewing key addr_{in} .*

For Monero, on the other hand, spend indistinguishability relative to the sole viewing key k^v seems to be “folklore” and taken for granted by the cryptocurrency community with little to no formal justification. It is implicitly assumed to reduce to *chosen-target linkable anonymity* of CLSAG ring signatures (see Definition 3 in Appendix B). It is, however, not clear how to, given only CLSAG public keys $\{(K_i^o, Z_i)\}_{i=1}^n$, simulate, in the reduction, the transaction T for the dictator $\mathcal{D}$. This is because of the directly observable algebraic structure of T . In Zcash and Firo, inputs and outputs of a transaction are shown to balance with a zero-knowledge argument, but, in Monero, a direct algebraic relation between the value commitments is checked by the validators which cannot be simulated in a polynomial-time reduction. Specifically, the reduction would need to sample (at least) three Pedersen commitments: C_{i_0} , C'_{i_1} , and C' that satisfy

$$C' = C_{i_0} - Z_{i_0} = C'_{i_1} - Z_{i_1},$$

and such that trapdoors of C_{i_0} and C'_{i_1} are known, as they get encrypted under k^v in u_{i_0} and u_{i_1} , respectively (see Appendix A.1). Once the viewing key is

⁷ See <https://zcash.github.io/orchard/design/nullifiers.html> (accessed: 30 August 2026).

disclosed, one cannot rely on pseudorandomness of encryption, since it is the *plaintexts* (now available to $\mathcal{D}$) that must be consistent with each other (see, however, Theorem 4 in the next section).

The same argument applies to any underlying ring signature (including the previously used MLSAG [46]). Indeed, it is easily seen that no black-box reduction is possible as long as the signing keys are derived from the openings of the value commitments, which must be disclosed to $\mathcal{D}$. The implicit belief of the Monero community (as well as a silent assumption in [25]) is thus merely reformulated below in the language of Definition 1.

Conjecture 1. Monero has indistinguishable spends relative to the incoming viewing key k^v .

4 General Viewing Key-Based Oversight

This next section studies how (in)effective viewing keys are at facilitating financial oversight of uncooperative subjects. As above, consider the practical setting (minimizing on-chain data), where subjects are supposed to submit transactions to an off-chain audit under legal obligation alone [10,25] (or to disassociate themselves from criminals [18]). Note that, for the case of an exchange or another “crypto-asset service provider” [71] undergoing an audit as well as for the setting of lightweight wallets offloading scanning responsibility to untrusted servers, it was appropriate to limit ourselves in Definition 1 to honestly minted UTXOs. This is not true when evaluating the effectiveness of viewing key-based oversight in general. We must instead consider a more elaborate experiment, where the notes coming into the subject’s possession may be formatted maliciously. In particular, their corresponding witnesses may be sent completely out of band.

4.1 Spend Indistinguishability Under Off-Chain Communication

Formally, one may think of the above setting as adding an extra “generation” of notes to the experiment in Definition 1, namely, running first an additional spender algorithm $\mathcal{S}'$ that produces transactions consistent with L that spend $u_1, \dots, u_n$ and result in a new UTXO set $u'_1, \dots, u'_n$ and a new ledger state L' . The experiment then proceeds as before with L' instead of L and with u'_i instead of the u_i , for each $i \in \{1, \dots, n\}$. $\mathcal{S}$ receives the owned UTXO $\bar{u}'_{i_b}$ as before, but now u_{i_0} and u_{i_1} are not guaranteed to contain witnesses encrypted under any viewing key in V . They may instead be “shadow UTXOs” whose witnesses are sent off-chain.

Definition 2. A *cryptocurrency* $\Pi = (\text{Setup}, \text{Coinbase}, \text{Spends})$ has indistinguishable spends relative to viewing keys V in the presence of out-of-band channels if there exist efficient spender algorithms $\mathcal{S}$ and $\mathcal{S}'$ such that the output of $\mathcal{S}'$ is computationally indistinguishable from the output of Coinbase without

any viewing keys, and, for all efficient dictators $\mathcal{D}$ and all anonymity set sizes $2 \leq n \leq \text{poly}(\lambda)$, there exists a negligible function ε such that, for all $b \in \{0, 1\}$,

$$\Pr[\text{SpendInd}_{\Pi, \mathcal{S}', \mathcal{S}, \mathcal{D}}^b(\lambda) = (\cdot, 1)] \geq 1 - \varepsilon(\lambda),$$

and

$$\left| \Pr[\text{SpendInd}_{\Pi, \mathcal{S}', \mathcal{S}, \mathcal{D}}^0(\lambda) = (1, \cdot)] - \Pr[\text{SpendInd}_{\Pi, \mathcal{S}', \mathcal{S}, \mathcal{D}}^1(\lambda) = (1, \cdot)] \right| \leq \varepsilon(\lambda),$$

where the experiment $\text{SpendInd}_{\Pi, \mathcal{A}, \mathcal{S}, \mathcal{D}}^b$ is presented in Figure 1, and the probabilities are taken over the random coins of all algorithms involved.

As a minor technicality, observe that we do not prevent $\mathcal{S}'$ from minting two notes with identical tags (tag non-collision is only verified by the network upon spending and not at the time of minting). Explicitly banning it is redundant, however, as a tag collision between $\bar{u}_{i_b}$ and $\bar{u}_{i_{1-b}}$ fails the compliance check and forces $g = 0$.

Theorem 3. *Zcash (resp. Firo) has indistinguishable spends relative to the full viewing key fvk (resp. $\text{addr}_{\text{full}}$) in the presence of out-of-band channels.*

Proof (sketch). The UTXOs $u_1, \dots, u_n$ that the dictator sees need not carry any information about their owners whatsoever. Indeed, the consistency of any ciphertexts published on chain is not verified by the consensus protocol, and so $\mathcal{S}'$ may encrypt random data under random keys. $\mathcal{S}$ nevertheless receives the witnesses for u_{i_b} out of band in the experiment. Even with the full viewing key fvk (resp. $\text{addr}_{\text{full}}$), the dictator is unable to compute the tags (see Equations (4) and (5) in Appendix A), which in fact appear random to the dictator. Indistinguishability of T spending one “shadow UTXO” u_{i_0} from T spending another “shadow UTXO” u_{i_1} then follows immediately from the zero-knowledge property of any underlying proof systems (see Appendix A.2 for Zcash and Appendix A.3 for Firo). $\square$

Theorem 4. *Monero has indistinguishable spends in the random oracle model relative to the incoming viewing key k^v in the presence of out-of-band channels.*

Proof (sketch). The statement follows from a reduction to chosen-target linkable anonymity of the CLSAG signature scheme (Definition 3). Since in the experiment above, the UTXOs $u_1, \dots, u_n$ are produced by $\mathcal{S}'$, not necessarily honestly, the reduction does not actually need to know the trapdoors to any commitments and can simulate them for $\mathcal{D}$ given just the CLSAG keys, which was not possible before (see discussion above). Specifically, for each CLSAG public key (K^o, Z) , we can simulate the corresponding “shadow UTXO” u by setting $u = (K^o, R, C, c)$ for R, C uniform in $\mathbb{G}$ and c a uniform string. The transaction

$$T = (\{u_1, \dots, u_k\}, C', \sigma, \tilde{K}, \tilde{Z}, \pi, u')$$

can then be simulated by first simulating the Bulletproofs+ [24] range proof π , setting $C' \leftarrow C - Z$ (see Appendix A.1), and obtaining the signature σ together with $\tilde{K}$ and $\tilde{Z}$ from the CLSAG oracle. $\square$

Note that, in the real world, no actual out-of-band communication needs to take place. Indeed, the witnesses could still be placed on chain but encrypted under an independent *double key* $dkey$ whose existence a subject can plausibly deny [48, 49, 59]. This “hidden in plain sight” approach reflects a real concern of criminals which is minimizing interactions with their accomplices that could leave traces for the law enforcement to follow [25, 44]. Here, they seem to only ever be interacting with the blockchain, no extra out-of-band traffic.

Furthermore, ideas from *anamorphic spending*, introduced in [25], could be used, whereby the subject, call her Alice, sends any witnesses to her accomplice, Bob, using subliminal channels afforded by the cryptocurrency itself [4, 13, 25]. For Monero, naturally, the biggest concern in this regard remains the vast covert bandwidth available in CLSAG signatures [39] recognized in [25] (see Appendix B). Surprisingly, however, what becomes a concern for Zcash and Firo is the “uncontrolled randomness” [4] in state-of-the-art zero-knowledge arguments. These may accommodate wide subliminal channels through the standard technique of *rejection sampling* alone [59]. For example, a typical 2-input-2-output transaction involves, due to proofs alone (*i.e.*, discounting signatures, value commitments, randomized public keys, *etc.*), 17 independently rerandomizable 256-bit elliptic curve points in Monero, 32 in Firo, and 77 in Zcash! Each such point P can be rejection-sampled until the evaluations $F_{dkey}(P)$ spell out “chunks” of some covert message m for a pseudorandom function F_{dkey} keyed with the double key $dkey$ shared between Alice and Bob. This m can contain any missing witnesses for notes already minted on chain or even entire transactions appropriately compressed to be broadcast by a proxy (*cf.* the no-proxy assumption underlying deanonymizations today [30]). In general, an entire underground payment network can be built that settles on the same public ledger, right under the nose of the dictator lulled into a false sense of security by mandating the disclosure of the full viewing key.

4.2 Conclusions and Remedy

What the previous section showed is that existing viewing keys are ineffective as a general auditability instrument when *all* monitored parties are malicious. Specifically, if a dictator were to demand the disclosure of all viewing keys for Monero, Zcash, and Firo today, their subjects could first “conceal” their coins (*e.g.*, during *vacatio legis*) by sending them back to themselves, this time formatting the notes like S' above would, namely, not encrypting any witnesses under any compromised viewing keys. The dictator would then have no way of recognizing the new “shadow UTXOs” even if the subjects truthfully surrendered their viewing keys. Fundamentally, this is because the consensus protocol is not concerned in any way with verifying the consistency of the ciphertexts, whereas viewing key-based oversight currently critically relies on these ciphertexts being consistent.

Let us now briefly sketch a maximally lightweight change to existing privacy coin protocols that would remedy this while minimizing on-chain data

(*cf.* [35, 69]). In doing so, we admittedly depart from the anamorphic cryptography setting, where we assume no changes to the underlying protocols are possible. It is, however, instructive to see how auditability can be facilitated with minor changes to consensus rules. Let a be a subject’s address and v the viewing key associated to a that has been surrendered to the dictator. The dictator wants an on-chain (*i.e.*, not requiring online interaction with any subject) oracle that, given a UTXO u and the key v , verifies if u is addressed to a . Note that, once ownership has been established this way, then the dictator can obligate the subject to attest to additional details of u as in [10, 18, 25]. To this end, suppose we extend every UTXO u to include an *attribution tag* A that enables offline spend attribution. Namely, instead of publishing u in the ledger, publish $\hat{u} = (u, A)$, where $A = (r, F_v(r), \pi)$, for uniformly random r , a pseudorandom function F keyed with the viewing key v , and a zero-knowledge argument π showing that v with which F was keyed is the same v that corresponds to the address a of the recipient of u . Pseudorandomness of F and zero knowledge of π then guarantee that privacy properties of the original cryptocurrency remain intact. At the same time, the dictator can use their knowledge of v to efficiently and, most importantly, in an offline fashion identify UTXOs in the possession of their subjects.

Three points are important to make. First, validity of π must be checked by the consensus protocol. This is the “anchor” that allows the dictator to trust the oracle. Second, the pseudorandom function $F_v(r)$ needs to have some algebraic structure to be computed by the sender who mints u . This is in general *not* the subject corresponding to the viewing key v (who is instead the recipient of u). In the case of Monero, for example, we could have $F_{k^v}(R) := k^v \cdot R$, which is a pseudorandom function under the Decisional Diffie-Hellman (DDH) assumption. The proof π would then correspond to the proof of equality of discrete logarithms (*cf.* **OutProof** in [46]). Finally, note that, even if we extended UTXOs with attribution tags, the dictator would still not be able to take for granted the consistency of the ciphertexts published on chain and would separately need to demand proofs about the structure of transactions [10, 18]. In particular, the dictator should have the relevant subject reveal the tag s of the (now correctly attributed) UTXO u and prove its consistency (*cf.* [25]). With this, the dictator would be able to recognize when this asset gets spent and request appropriate attestations and disclosure at that point also. The auditing functionality would then finally be sound, in that no transaction would go unaccounted for: inbound transfers would be identified by their attribution tags A and outbound transfers by their (regular) tags s . Refusal to submit a transaction to an audit could be prosecuted accordingly (*cf.* [22, Insight 9]). This is of particular importance when dealing with institutional exposure to privacy coins, which, *e.g.*, the AML regulation in Europe currently aims to strictly prohibit (see [72, Article 79(1)]).

In closing, observe that, in many practical settings, the stronger model of Definition 1 may be in fact be more appropriate. For example, during a withdrawal to a self-hosted wallet (which is paid extra attention by the regulators; see, *e.g.*, [68] or [70, Article 14(5)]), it is reasonable to assume that a licensed

exchange would mint the UTXOs for the wallet honestly. The simpler security model of Definition 1 then accurately describes monitoring the client afterwards.

5 Summary and Closing Remarks

Despite the European Court for Human Rights’ establishing in a landmark ruling [34] that

[permitting] the public authorities to have access, on a generalised basis and without sufficient safeguards, to the content of electronic communications . . . impairs the very essence of the right to respect for private life[,]

it seems no such principle applies to privacy in one’s economic life, and the legal crackdown on anonymity-enhancing cryptocurrencies intensifies. Indeed, cryptocurrencies have become a new front of the so-called “crypto wars” [1, 2], with legislators willing to sacrifice financial privacy on the altar of accountability and transparency. To not risk their banning privacy coins for good in the conviction they are nothing more than laundromats [5], it becomes crucial to identify solid mechanisms for oversight that strike a balance between lawful access and data minimization [69], particularly in the age of AI-powered financial surveillance. Viewing keys are often proposed as a means of addressing this issue [26, 27, 33, 58], but the security around them so far lacked systematic evaluation.

The present paper has laid the groundwork for evaluating both existing solutions, namely, Zcash, Monero, and Firo, as well as future designs, such as, *e.g.*, the upcoming Monero releases: Seraphis and Jamtis, by introducing two models of spend indistinguishability relative to viewing keys (Definitions 1 and 2). In the latter model, we showed that *all* viewing keys defined today for mainstream privacy coins are ineffective against sophisticated criminals determined to evade AML monitoring. Indeed, they may happily comply with any demand to surrender even the most powerful viewing keys and nonetheless continue to run covert operations on the public ledger itself (Theorems 3 and 4). Notably, however, while not a bulletproof solution, viewing keys have their place in a risk-based approach to combating, say, money laundering or financing of terrorism through institutional actors such as exchanges (*cf.* the exemption of person-to-person transfers in [70, Article 2(4)]), where the stronger model of Definition 1 may apply better. We also filled in the gaps in provable anonymity of Zcash and Firo in a more moderate setting, where only the incoming viewing keys are compromised (Theorems 1 and 2), and identified flaws in the design and/or security assumptions of Monero that hinder provably establishing spend indistinguishability relative to its sole viewing key despite it being assumed already by its users. Resolving Conjecture 1 is thus an urgent next step. It entails one of the following: changes to the Monero protocol, reformulating CLSAG [39] security in the language of zero knowledge and simulatability, or doing a white-box reduction to the DDH problem directly. This is particularly important given that incoming viewing keys find everyday use in offloading blockchain scanning to third parties [75]. Finally, the cautionary tale told by Section 4.1 should be addressed. In

particular, the vast covert bandwidth afforded by state-of-the-art zero-knowledge arguments makes using them in public risky at best (*cf.* also [13]), an irony, given how instrumental the cryptocurrency boom was to sparking interest in such technologies. Principles and techniques developed recently by Quach *et al.* [62] can be leveraged to remedy this.

In closing, note that, in the setting of an authoritarian regime, the findings of this work are clearly good news, protecting journalists, activists, and NGOs from financial surveillance and censorship. On the other hand, whether they are decidedly bad news or good news is not clear in a democratic society. As always, technology empowers indiscriminately, regardless of intentions, and whether the risks outweigh the benefits remains an open question.

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A Monero, Zcash, and Firo in More Detail

This section gives a more comprehensive introduction to the most relevant privacy coins today, namely, Monero and Zcash, as well as Firo, which is of theoretical interest to the present work.

A.1 Monero

Monero [46], based on CryptoNote [65], is one of the earliest privacy coins to gain mainstream recognition. Its notes have the form:

$$u = (K^o, R, C, c),$$

with $K^o, R, C \in \mathbb{G}$ and c a ciphertext. Users are identified by their public addresses $(K^v, K^s) \in \mathbb{G}^2$. To mint a UTXO u addressed to (K^v, K^s) , the originator

should sample $r \leftarrow \mathbb{F}_q$, set $R \leftarrow r \cdot G$ and

$$K^o \leftarrow K^s + \text{Hash}_{\mathbb{F}_q}(r \cdot K^v) \cdot G.$$

We call K^o the (one-time) *stealth address*. The recipient knows the secret spending key k^s and the secret viewing key k^v such that, respectively, $K^s = k^s \cdot G$ and $K^v = k^v \cdot G$. They are therefore able to compute k^o such that $K^o = k^o \cdot G$ by setting

$$k^o \leftarrow k^s + \text{Hash}_{\mathbb{F}_q}(k^v \cdot R).$$

This *output key* is used to authorize the spending of u . C is supposed to be a Pedersen commitment to the amount v attached to u . Specifically, the sender should set

$$C \leftarrow x \cdot G + v \cdot H,$$

for some trapdoor x . (Recall that G and H are two generators of $\mathbb{G}$ sampled independently.) The sender should encrypt the witnesses x and v under the Diffie-Hellman key $r \cdot K^v$ and set c to this ciphertext [46]:

$$c \leftarrow \text{Enc}_{r \cdot K^v}(x, v).$$

We stress that this is merely prescribed by the specification of Monero, but not enforced by its consensus protocol, hence “should” as opposed to “must.” Note how, under normal circumstances, where users adhere to the specification, the viewing key k^v enables the decryption of notes without granting spend authority.

Balance. To prove balance, *i.e.*, that the net output amount equals the net input amount,⁸ Monero transactions rely on Pedersen commitments’ being homomorphic. Let $C_j = x_j \cdot G + v_j \cdot H$ be the value commitment of the j th input note. For each input note, a new commitment C'_j (referred to as a *pseudo-output commitment* in [46]) is constructed that commits to the same amount v_j but uses a different trapdoor x'_j . The value commitments $\hat{C}_t$ of the outputs (newly minted notes) are then constructed as follows. The trapdoors y_t are selected so that $\sum_j x'_j - \sum_t y_t = 0 \pmod{q}$. If the input and output amounts balance, then the sum of the pseudo-output commitments will be equal to the sum of the output commitments:

$$\sum_j C'_j - \sum_t \hat{C}_t = \infty,$$

where ∞ denotes the neutral element of $\mathbb{G}$. This can be publicly checked by the network at large. This design choice, where a direct algebraic relation is verified instead of, say, a zero-knowledge argument has consequences for the provability of the anonymity of Monero (see Conjecture 1). Notably, the above implication only goes one way. Sums of commitments being equal only shows that amounts balance modulo q and not necessarily over $\mathbb{Z}$. To show that no

⁸ Miner’s fees are neglected here and in the rest of the paper. They are straightforward to account for.

overflow has occurred, output amounts are also proved to lie in the correct range using a Bulletproofs+ range proof π [24]. What remains to be shown is that pseudo-output commitments are consistent with the input commitments. This is proved by showing, for each input j , the knowledge of $z_j = x_j - x'_j$ such that $C_j - C'_j = z_j \cdot G$.

Tags and Authorization. Both the knowledge of z_j and of the output key k_j^o are proved together with a CLSAG [39] linkable ring signature. Using ring signatures affords the sender k -anonymity, whereby outside observers cannot in general identify which UTXO was really spent with probability better than $1/k$.⁹

CLSAG (as used in Monero) has private keys of the form $(k^o, z) \in \mathbb{F}_q^2$ and public keys $(k^o \cdot G, z \cdot G) \in \mathbb{G}^2$.¹⁰ The j th input to a transaction gets signed with (k_j^o, z_j) , where k_j^o is the output key of the input note, and z_j is such that $z_j \cdot G = C_j - C'_j$. Note that $C_j - C'_j$ is supposed to be a commitment to the value zero (the H -components cancel). z_j is then its opening. To hide the true input, $k-1$ *decoy* inputs are chosen at random from the ledger. These are, in general, other users' notes (possibly even already spent). Their stealth addresses K_{ji}^o and output commitments C_{ji} are used in the CLSAG signature. Specifically, the decoy public keys become $\{(K_{ji}^o, C_{ji} - C'_j)\}_{i=1}^{k-1}$. The real public key $(K_j^o, C_j - C'_j)$ is then shuffled in at some random position $\ell \in \{1, \dots, k\}$. The signature only shows that the transaction originator knows the secret key corresponding to *one* of the public keys involved. No efficient adversary can tell with non-negligible advantage over $1/k$ which public key was actually used to produce the signature [39]. Two *key images* are involved in the signature and revealed alongside it: a primary key image $\tilde{K} = k_j^o \cdot \text{Hash}_{\mathbb{G}}(K_j^o)$ and an auxiliary key image $\tilde{Z} = z_j \cdot \text{Hash}_{\mathbb{G}}(K_j^o)$ [46]. The primary key image $\tilde{K}$ plays the role of the note's tag: if double-spending is attempted, $\tilde{K}$ will be repeated, and thus reuse of k^o will be detected.

A more detailed description of the CLSAG signature scheme can be found in Appendix B. For now, let us summarize what a Monero transaction looks like. For simplicity, consider a single-input single-output transaction T that spends a UTXO $u \in \{u_1, \dots, u_k\}$ and mints u' . Then

$$T = (\{u_1, \dots, u_k\}, C', \sigma, \tilde{K}, \tilde{Z}, \pi, u'),$$

with C' a pseudo-output commitment, σ a CLSAG ring signature [39], $\tilde{K}, \tilde{Z} \in \mathbb{G}$ the two key images, and π a Bulletproofs+ [24] range proof.

⁹ With $k = 16$ as of this writing. See <https://xmrchain.net> (accessed: 30 August 2026). Notably, this simplified view ignores correlations between transactions that increase the distinguisher's advantage when the *cover set* is not chosen carefully [32, 73, 74]. This is not relevant for the present work.

¹⁰ Goodell *et al.* [39] define CLSAG more generally for public keys $(Z_1, \dots, Z_d) \in \mathbb{G}^d$ for some d . What is used in Monero corresponds to 2-CLSAG using their terminology.

A.2 Zcash

Zcash is based on Zerocash [11], itself based on Zerocoin [52]. The present work focuses solely on Zcash’s most recent *shielded protocol*,¹¹ which, as of this writing, is *Orchard* [40].¹² An Orchard note has a particularly simple form:

$$u = (\text{cm}, \text{epk}, C^{\text{enc}}),$$

where cm is a *note commitment* and the pair $(\text{epk}, C^{\text{enc}})$ is supposed to be a ciphertext in an ElGamal-based hybrid encryption scheme carrying the witnesses for the recipient.¹³ The note commitment cm is to be computed as

$$\text{cm} \leftarrow \text{Hash}_{\mathbb{G}}(\mathbf{g}_d, \mathbf{pk}_d, v, \rho, \psi) + \text{rcm} \cdot H,$$

where $\mathbf{g}_d$ and $\mathbf{pk}_d$ are components of the recipient’s address, v is the note’s amount, rcm and ψ are randomizers sampled by the sender, and ρ is the tag of one of the inputs to the transaction that minted the note (see below). When a note u is minted, its commitment cm is added as a leaf to a global append-only Merkle tree, called the *note commitment tree* [40, Section 3.8]. To spend u , one must prove that cm is present in the tree, that one knows the secret spending key associated to $\mathbf{g}_d$ and $\mathbf{pk}_d$, and that the amounts in the transaction balance overall. All this is proved in zero knowledge meaning in particular that the anonymity set for each spend is equal to at least the set of all outstanding UTXOs. Notably, this set is itself implicit as one cannot in general decide if a given note has already been spent or not.

Key Hierarchy. Let us now elaborate the simple idea from above, starting with the key hierarchy defined by Orchard. At the top of this hierarchy, we find the *spend authorizing key* $\text{ask} \in \mathbb{F}_q$ and the *full viewing key*

$$\text{fvk} = (\text{ak}, \text{nk}, \text{rivk}) \in \mathbb{G} \times \mathbb{F}_p \times \mathbb{F}_q.$$

In practice, ask , nk , and rivk are typically derived from a single master seed. For simplicity, we will assume they are sampled independently. The *spend validating key* ak is derived as $\text{ak} \leftarrow \text{ask} \cdot G$. From the full viewing key one then derives three subsidiary viewing keys, namely, the *diversifier key* dk , the *incoming viewing key* ivk , and the *outgoing viewing key* ovk as follows.¹⁴ The incoming viewing key is derived as

$$\text{ivk} \leftarrow \mathbf{x}(\text{Hash}_{\mathbb{G}}(\text{ak}, \text{nk}) + \text{rivk} \cdot H) \bmod q. \quad (1)$$

¹¹ Zcash also supports a transparent protocol alongside it, which is equivalent to Bitcoin and hence of no interest.

¹² On 28 July 2026, an upgrade called *Ironwood* launched that introduced a new “shielded pool” but it nevertheless continues to use the Orchard protocol.

¹³ Analogous to (R, c) for Monero.

¹⁴ The Zcash specification [40] actually refers to the pair (dk, ivk) as the incoming viewing key. Since we will typically not be interested in dk , we shall for brevity reserve the term for ivk alone. This is also in line with the naming used in older versions of the Zcash protocol (Sapling).

In Zcash, we assume $q > p$. Note that ivk is computed as an x -coordinate of an element of $\mathbb{G}$, itself an element of the base field $\mathbb{F}_p$ (here identified with $\mathbb{Z}/(p)$), and only then lifted to the (larger) scalar field $\mathbb{F}_q$. This is of some minor significance in the proof of Theorem 1. The outgoing viewing key ovk and the diversifier key dk are derived jointly as

$$\text{dk} \parallel \text{ovk} \leftarrow \text{Hash}_{\{0,1\}^{4\lambda}}(\text{rivk}, \text{ak}, \text{nk}), \quad (2)$$

where dk and ovk are each 2λ bits long, two halves of the output of the hash function.

Addressing Notes. From (dk, ivk) one finally derives a payment address. In fact, many so-called *diversified addresses* can be derived from a single (dk, ivk) pair. To create a new address, a user picks an index $i \in \{0, 1\}^{88}$ and applies a pseudorandom permutation Enc keyed with dk to it to get a *diversifier*

$$d \leftarrow \text{Enc}_{\text{dk}}(i).$$

(We write Enc for the permutation, as it is implemented with the AES block cipher.) From the diversifier, a fresh generator of the group $\mathbb{G}$ is derived as

$$\mathbf{g}_d \leftarrow \text{Hash}_{\mathbb{G}}(d).$$

The diversified address corresponding to the index i is then the pair (d, pk_d) , where

$$\text{pk}_d = \text{ivk} \cdot \mathbf{g}_d.$$

Recall that a note u gets addressed to (d, pk_d) by committing to $\mathbf{g}_d$ and pk_d in the note commitment cm . Additionally, one normally encrypts the witnesses needed later to spend u , such as the randomizers rcm and ψ under the Diffie-Hellman key $\text{esk} \cdot \text{pk}_d$ for uniform $\text{esk} \leftarrow \mathbb{F}_q$, setting $\text{epk} \leftarrow \text{esk} \cdot \mathbf{g}_d$. This is the ciphertext C^{enc} attached to u .

Clearly, when u is minted honestly (*i.e.*, according to the specification), knowledge of ivk suffices to decrypt it. Consistency of C^{enc} is, however, not subject to the consensus protocol. The sender may choose to encrypt random data under a random key instead and only send the witnesses $(\psi, \text{rcm}, v, \text{etc.})$ out of band to the recipient [40, Section 4.20.1]. The holder of ivk in this case will not even be able to recognize that u credits pk_d . Indeed, incoming notes are normally identified solely based on whether they successfully decrypt under ivk or not, which such “malformed” u would not. Interestingly, an older version of the Zcash protocol, called Sapling, required the sender prove the ephemeral key epk was derived honestly, namely, prove they know esk [40]. Consistency of the ciphertext C^{enc} , however, was not verified.

When minting u , the sender may also choose to enable the holder of *their* (not the recipient’s) outgoing viewing key ovk to see the details of u . This is simply done by encrypting $(\text{pk}_d, \text{esk})$ under ovk . The corresponding ciphertext, denoted C^{out} , is included in the transaction. This is again done solely at the sender’s discretion [40, Section 4.20.1].

Actions. For the purposes of proving balance as well as tag derivation, the concept of *actions* will be important. An *action* is simply a spend-mint pair $(u_i, \hat{u}_j)$, where the note u_i is getting spent and, in the same transaction, the note $\hat{u}_j$ is getting minted. When creating a transaction, its originator must decide on how to match spends and mints. This is typically done at random.

For example, suppose we want to, in a single transaction, spend the UTXOs u_1, u_2 , and u_3 , and mint two new notes, say, $\hat{u}_1$ and $\hat{u}_2$. This is the very common scenario of aggregating funds from several UTXOs, here three, in order to make a payment in $\hat{u}_1$ and then send back change in $\hat{u}_2$. Since Zcash requires that spends and mints be paired together, and we are short one spend, we add a dummy one, call it $\hat{u}_3$, with value $v = 0$ addressed to a random recipient.¹⁵ We then choose a random matching, *e.g.*, $\{(u_1, \hat{u}_3), (u_2, \hat{u}_2), (u_3, \hat{u}_1)\}$.

Balance. When proving balance, the transaction originator attaches a *value commitment* cv^{net} to each action $(u, \hat{u})$. Let v be the amount attached to the input note u and let $\hat{v}$ be the amount attached to the output note $\hat{u}$. The transaction creator should sample the value commitment as

$$\text{cv}^{\text{net}} \leftarrow \text{rcv} \cdot G + (v - \hat{v}) \cdot H, \quad (3)$$

for a random trapdoor $\text{rcv} \leftarrow \$ \mathbb{F}_q$.

Let $\text{cv}_j^{\text{net}} = \text{rcv}_j \cdot G + (v_j - \hat{v}_j) \cdot H$ be the value commitment of the j th action and consider summing the commitments together:

$$\sum_j \text{cv}_j^{\text{net}} = \left(\sum_j \text{rcv}_j \right) \cdot G + \left(\sum_j v_j - \sum_j \hat{v}_j \right) \cdot H.$$

If the transaction balances, *i.e.*, $\sum_j v_j = \sum_j \hat{v}_j$, then the H -component above will vanish, in which case the originator will know the discrete logarithm of $\sum_j \text{cv}_j^{\text{net}}$ to base G , namely, $\sum_j \text{rcv}_j$. This knowledge is proved by signing the transaction to produce the so-called *binding signature* σ_{binding} [40, Section 4.14].¹⁶ Compare this with Monero, where the commitment trapdoors are sampled in a special way, so that the commitments sum to the identity element ∞ (*i.e.*, both the G - and H -components vanish). This direct algebraic relation is what causes problems when trying to prove Conjecture 1.

Tags and Authorization. When spending a note u , the transaction must reveal its tag (referred to as a *nullifier* in [40] or as a *serial number* in [11]). It

¹⁵ Dummy spends are also possible, see [40, Section 4.8.3] for details.

¹⁶ We note here a complication one finds in the real world that is purposefully overlooked here. As per an earlier footnote, Zcash supports a transparent protocol alongside Orchard. When sending funds from the transparent pool to the Orchard/Ironwood pool (shielding) or *vice versa* (desielding), one accounts for the public net change of supply v^{balance} in the Orchard/Ironwood pool with an extra commitment $v^{\text{balance}} \cdot H$.

is computed as follows:

$$\text{nf} \leftarrow \mathbf{x}((\text{Hash}_{\mathbb{F}_p}(\text{nk}, \rho) + \psi \bmod p) \cdot K + \text{cm}), \quad (4)$$

with K one of the fixed generators of $\mathbb{G}$, with $\psi \in \mathbb{F}_p$ committed to in u 's note commitment cm , $\rho \in \mathbb{F}_p$ fixed to be the tag of the note spent in the action that minted u , and nk the *nullifier deriving key* being part of the full viewing key fvk [40, Section 4.16].

To authorize spending u , one first randomizes the spend authorizing key ask by sampling $\alpha \leftarrow \$ \mathbb{F}_q$ and setting

$$\text{rsk} \leftarrow \text{ask} + \alpha.$$

One then signs the transaction with rsk and reveals the *randomized spend validating key* $\text{rk} = \text{rsk} \cdot G$ [40, Section 4.15]. The link between rk and the address pk_d committed to in cm is proved as part of the zk-SNARK π attached to the transaction and elaborated below.

Finally, everything is tied together with a Halo2 [16] zk-SNARK π that proves, among some other minor things, that, for every action $(u, \hat{u})$,

- the randomized spend validating key rk was correctly derived from ak , which in turn was used to derive ivk via Equation (1),
- the value commitment cv^{net} was correctly computed as per Equation (3),
- the input amount v and the output amount $\hat{v}$ lie within the correct range and do not lead to an overflow in cv^{net} ,
- the new note commitment is correct,
- the nullifier nf was computed correctly as per Equation (4),
- the old note commitment cm corresponds to an address $\text{pk}_d = \text{ivk} \cdot \mathbf{g}_d$, and
- cm was added in the past to the note commitment tree.

See [40, Section 4.18.4] for the full statement.

In summary, a (single-input single-output) Zcash transaction that spends a UTXO u and mints a new one u' looks like:

$$T = (\text{rt}, \text{cv}^{\text{net}}, C^{\text{out}}, \text{nf}, \sigma, \text{rk}, \sigma_{\text{binding}}, \pi, u'),$$

with rt the *anchor* corresponding to the root of the note commitment tree at some point in the past, cv^{net} the value commitment, C^{out} the “outgoing” ciphertext, nf the tag of the input note u , σ a spend authorizing signature under a randomized spend validating key rk , π a Halo2 [16] zk-SNARK, and σ_{binding} a binding signature.

A.3 Firo

Firo is based on Lelantus Spark [43] which draws heavily on both Monero and Zcash for its design. Indeed, a Firo UTXO looks almost identical to a Monero UTXO:

$$u = (S, R, C, c),$$

where $S, R, C \in \mathbb{G}$ and c is a ciphertext. As in Monero, C is a value commitment, *i.e.*, a Pedersen commitment to the amount attached to u , and R is an ephemeral key that is supposed to be used to decrypt c . S , on the other hand, is a commitment to u 's serial number s . When spending u , the transaction will need to reveal a tag derived from s .

Key Hierarchy. Firo uses a similar key hierarchy to Zcash. At the top we find the the spend authorizing key

$$\mathbf{addr}_{\text{sk}} = (s_1, s_2, r) \in \mathbb{F}_q^3,$$

sampled uniformly. From it, one derives the *full viewing key*

$$\mathbf{addr}_{\text{full}} \leftarrow (s_1, s_2, D, P_2) \in \mathbb{F}_q^2 \times \mathbb{G}^2,$$

where $D = r \cdot G$ and $P_2 = s_2 \cdot F + D$. From $\mathbf{addr}_{\text{full}}$, finally, one derives the *incoming viewing key*

$$\mathbf{addr}_{\text{in}} \leftarrow (s_1, P_2) \in \mathbb{F}_q \times \mathbb{G}.$$

No outgoing viewing key is defined.

Addressing Notes. From the incoming viewing key one derives diversified addresses exactly as in Zcash. To create a new address, a user picks an index $i \in \mathbb{N}$ and encrypts it with s_1 to get a diversifier¹⁷

$$d \leftarrow \text{Enc}_{s_1}(i).$$

The diversifier then determines a fresh generator of $\mathbb{G}$ exactly as in Zcash. One sets

$$Q_1 \leftarrow s_1 \cdot \text{Hash}_{\mathbb{G}}(d) \quad \text{and} \quad Q_2 \leftarrow \text{Hash}_{\mathbb{F}_q}(2, s_1, i) \cdot F + P_2.$$

The diversified address corresponding to the index i is then the tuple (d, Q_1, Q_2) .

To send a note u to (d, Q_1, Q_2) , the sender samples a nonce $z \leftarrow \$ \mathbb{F}_q$ and commits to a serial number by setting

$$S \leftarrow \text{Hash}_{\mathbb{F}_q}(3, z) \cdot F + Q_2.$$

The recipient, given the nonce z , the diversifier d , and their full viewing key $\mathbf{addr}_{\text{full}}$ can compute this serial number as

$$s \leftarrow \text{Hash}_{\mathbb{F}_q}(3, z) + \text{Hash}_{\mathbb{F}_q}(2, s_1, i) + s_2,$$

where they find the index as $i \leftarrow \text{Dec}_{s_1}(d)$ [43, Section 4.7]. Observe that

$$S = s \cdot F + D.$$

¹⁷ Jivanyan and Feickert [43] call i itself a diversifier, but this work shall adhere to the naming convention established by Zcash [40].

The value commitment trapdoor is derived from the nonce z . Specifically, the sender sets

$$C \leftarrow v \cdot G + \text{Hash}_{\mathbb{F}_q}(4, z) \cdot H.$$

The component Q_1 of the address is only used for the in-band distribution of witnesses. Specifically, the nonce z , the diversifier d , and the amount v attached to u are normally supposed to be encrypted (using again an ElGamal-based hybrid encryption scheme) and attached to the note. Indeed, the sender should compute an ephemeral key

$$R \leftarrow \text{Hash}_{\mathbb{F}_q}(1, z) \cdot \text{Hash}_{\mathbb{G}}(d),$$

and encrypt the note under $E \leftarrow \text{Hash}_{\mathbb{G}}(1, z) \cdot Q_1$:

$$c \leftarrow \text{Enc}_E(v, d, z).$$

Clearly, when u is minted honestly like this, then the knowledge of addr_{in} (indeed, s_1) suffices to decrypt it.

As in the case of Zcash and Monero, the consistency of the ciphertext c is not verified in any way by the consensus protocol. Also, successful decryption of c is the only way the holder of addr_{in} can identify that the note credits (d, Q_1, Q_2) . If c is malformed and witnesses, say, z are sent out of band, then the holder of addr_{in} is completely oblivious as to whom u credits.

Sender Anonymity. To hide which UTXO u is being spent, Firo uses a technique similar to Monero, namely, the sender pulls in $k - 1$ decoys at random from the ledger and demonstrates that one out of the k UTXOs referenced is being spent. To this end, the sender includes in a transaction two “commitment offsets” [43, Section 4.8]: a *serial number commitment offset*

$$S' \leftarrow s \cdot F - \text{Hash}_{\mathbb{F}_q}(5, s, D) \cdot H + D$$

and a *value commitment offset*

$$C' \leftarrow v \cdot G + \text{Hash}_{\mathbb{F}_q}(6, s, D) \cdot H,$$

where s and v are, respectively, the serial number and the amount associated with u , and D is the component of the full viewing key $\text{addr}_{\text{full}}$.

Their purpose is analogous to Monero’s pseudo-output commitment (see Appendix A.1). Recall that the serial number commitment S of u satisfies $S = s \cdot F + D$. Therefore, the sender is expected to know the discrete logarithm of $S - S'$ to base H , which is $\text{Hash}_{\mathbb{F}_q}(5, s, D)$. Similarly, the sender should know the discrete logarithm of $C - C'$ to base H . They demonstrate the knowledge of both using a *parallel one-out-of-many proof* [43, Section 2.4] for the relation

$$\left\{ \{S_i, V_i\}_{i=1}^k \subseteq \mathbb{G}^2, S', V' \in \mathbb{G}; \ell \in \mathbb{N}, s, v \in \mathbb{F}_q \mid \right. \\ \left. 1 \leq \ell \leq k \text{ and } S_\ell - S' = s \cdot H \text{ and } V_\ell - V' = v \cdot H \right\}.$$

Note how the proof is functionally equivalent to the CLSAG ring signature from Monero (see Appendix B), in that it shows the prover knows the openings of the two commitments for some note in the cover set $\{u_1, \dots, u_k\}$. Notably, however, because it is formulated as a non-interactive zero-knowledge proof, one can invoke its simulator to prove indistinguishability, which is not possible in the case of CLSAG (see Conjecture 1).

Balance. Balance is proved analogously to how it is done in Zcash, namely, the sender demonstrates the knowledge of the discrete logarithm of the aggregate value commitment $\sum_j C_j - \sum_t \hat{C}_t$ to base H , where C_j is the value commitment of the j th input and $\hat{C}_t$ is the value commitment of the t th output. In doing so, they demonstrate that the G -component has vanished, *i.e.*, the amounts balance modulo the group order. (Compare again with Monero, where both components vanish.) This is then coupled with a simple *range proof* [43, Section 2.7] to account for the possibility of overflow.

Tags and Authorization. To spend u , its owner computes a tag

$$N \leftarrow \frac{1}{s} \cdot (K - D), \quad (5)$$

where s is u 's serial number, D is the component of the full viewing key $\text{addr}_{\text{full}}$, and K is a fixed generator of $\mathbb{G}$. They reveal N and prove that it is correctly computed, namely, that it is consistent with the serial number commitment offset S' . They also prove to know r such that $D = r \cdot G$. This is achieved with a *modified Chaum-Pedersen proof* [43, Section 2.3] for the relation

$$\begin{aligned} & \{ \{S'_j, N_j\}_{j=1}^w \subseteq \mathbb{G}^2; \{x_j, y_j, z_j\}_{j=1}^w \subseteq \mathbb{F}_q^3 \mid \\ & (\forall j \in \{1, \dots, w\})(S'_j = x_j \cdot F + y_j \cdot G + z_j \cdot H \text{ and } K = x_j \cdot N_j + y_j \cdot G) \}, \end{aligned}$$

which batches the above claims across all w inputs to a transaction.

In summary, a single-input single-output Firo transaction that spends a UTXO $u \in \{u_1, \dots, u_k\}$ and mints u' has the format:

$$T = (\{u_1, \dots, u_k\}, S', C', N, \pi_{\text{par}}, \pi_{\text{chaum}}, \pi_{\text{bal}}, \pi_{\text{rp}}, u'),$$

where S' and C' are the commitment offsets defined above, N is the input note u 's tag, π_{par} is the parallel one-out-of-many proof showing that S' and C' are consistent with some u_ℓ , π_{chaum} is the modified Chaum-Pedersen proof showing consistency of N and spend authority (knowledge of r), π_{bal} is a Schnorr proof demonstrating balance, and π_{rp} is range proof showing that the amount attached to u' is in the correct range.

B CLSAG in Monero

This section elaborates the CLSAG ring signature scheme as used in Monero.

Consider a Monero transaction T and focus on its j th input denoted u . Let $\{u_1^{(j)}, \dots, u_k^{(j)}\}$ be the cover set for this input with the real input at position ℓ . This means that $u_\ell^{(j)} = u$ and the remaining $u_i^{(j)}$ for $i \neq \ell$ were sampled from the Monero ledger. Parse each note in the cover set as

$$u_i^{(j)} = (K_{ji}^o, R_{ji}, C_{ji}, c_{ji}).$$

The originator of T samples a pseudo-output commitment C'_j for this j th input as described in Appendix A.1, so that they know z_j such that $z_j \cdot G = C_{j\ell} - C'_j$. The CLSAG private key becomes $(k_j^o, z_j) \in \mathbb{F}_q^2$, where k_j^o is the output key such that $K_{j\ell}^o = k_j^o \cdot G$. Write $Z_{ji} = C_{ji} - C'_j$. The decoy ring becomes $\{(K_{ji}^o, Z_{ji})\}_{i=1}^k$.

The CLSAG signing and verification procedures are presented in Figures 2 and 3, respectively. The key generation algorithm is straightforward and omitted for brevity: it samples k^o and z uniformly from $\mathbb{F}_q$ and returns $\text{sk} = (k^o, z)$ and $\text{pk} = (k^o \cdot G, z \cdot G)$. Observe in Figure 2 that, out of the k nonces $r_1, \dots, r_k \in \mathbb{F}_q$ comprising a CLSAG signature, only one is derived deterministically, namely, r_ℓ . The remaining ones are drawn uniformly at random from $\mathbb{F}_q$. This is the basis for the subliminal channel studied in [25] and alluded to in Section 4.1.

For completeness, let us recall the *linkable anonymity* notion that CLSAG signatures are shown in [39] to satisfy. It is implicitly assumed by the community that this implies spend indistinguishability for Monero (Conjecture 1). As explained in Section 3, however, this is not the case.¹⁸

Definition 3 (Chosen-Target Linkable Anonymity). *A linkable ring signature scheme $\Sigma = (\text{KeyGen}, \text{Sign}, \text{Verify})$ is linkably anonymous if, for all efficient algorithms $\mathcal{A}$ and all k , there exists a negligible function ε such that*

$$\Pr[\text{LinkAnon}_{\mathcal{A}, \Sigma}^k(\lambda) = 1] \leq \varepsilon(\lambda),$$

where the experiment $\text{LinkAnon}_{\mathcal{A}, \Sigma}^k$ is presented in Figure 4, and the probability is taken over the random coin tosses of all algorithms involved.

C Proofs

This section revisits the claims about Zcash and Firo having indistinguishable spends relative to incoming viewing keys and formally establishes Theorems 1 and 2. Both proofs proceed in a similar fashion, using a standard technique of defining a sequence of hybrids starting with $G_0 = \text{SpendInd}_{H, \text{Coinbase}, S, \mathcal{D}}^0$ and ending with $\text{SpendInd}_{H, \text{Coinbase}, S, \mathcal{D}}^1$. Each consecutive pair of games will be indistinguishable to efficient adversaries thereby “interpolating” between the two extremes and proving them indistinguishable. Write p_i for the probability that $\mathcal{D}$ returns 1 in the i th hybrid G_i and write, for brevity, $G_i \mapsto G_j$ to denote the

¹⁸ One could, notably, try reducing spend indistinguishability of Monero to the underlying DDH problem directly (a white-box reduction “through” CLSAG).

```

Sign(sk, m, {pk1, ..., pkk})
1 : Let  $\ell$  be such that  $\text{sk} \cdot G = \text{pk}_\ell$ 
2 :  $(k_j^o, z_j) \leftarrow \text{sk}$ 
3 :  $Q \leftarrow \text{pk}_1 \parallel \dots \parallel \text{pk}_k$ 
4 : foreach  $i \in \{1, \dots, k\}$  do
5 :    $(K_{ji}^o, Z_{ji}) \leftarrow \text{pk}_i$ 
6 : endforeach
7 :  $\tilde{K} \leftarrow k_j^o \cdot \text{Hash}_{\mathbb{G}}(K_{j\ell}^o)$ 
8 :  $\tilde{Z} \leftarrow z_j \cdot \text{Hash}_{\mathbb{G}}(K_{j\ell}^o)$ 
9 : foreach  $i \in \{1, \dots, k\} \setminus \{\ell\}$  do
10 :    $r_i \leftarrow \mathbb{F}_q$ 
11 :    $W_i \leftarrow \text{Hash}_{\mathbb{F}_q}(1, Q, \tilde{K}, \tilde{Z}) \cdot K_{ji}^o + \text{Hash}_{\mathbb{F}_q}(2, Q, \tilde{K}, \tilde{Z}) \cdot Z_{ji}$ 
12 : endforeach
13 :  $w_\ell \leftarrow \text{Hash}_{\mathbb{F}_q}(1, Q, \tilde{K}, \tilde{Z}) \cdot k_j^o + \text{Hash}_{\mathbb{F}_q}(2, Q, \tilde{K}, \tilde{Z}) \cdot z_j$ 
14 :  $\tilde{W} \leftarrow w_\ell \cdot \text{Hash}_{\mathbb{G}}(K_{j\ell}^o)$ 
15 :  $\alpha \leftarrow \mathbb{F}_q$ 
16 :  $c_{(\ell \bmod k)+1} \leftarrow \text{Hash}_{\mathbb{F}_q}(Q, m, \alpha \cdot G, \alpha \cdot \text{Hash}_{\mathbb{G}}(K_{j\ell}^o))$ 
17 : for  $i = (\ell \bmod k) + 1, \dots, (\ell - 1 \bmod k) + 1$  do
18 :    $c_{(i \bmod k)+1} \leftarrow \text{Hash}_{\mathbb{F}_q}(Q, m, (r_i \cdot G + c_i \cdot W_i), (r_i \cdot \text{Hash}_{\mathbb{G}}(K_{ji}^o) + c_i \cdot \tilde{W}))$ 
19 : endfor
20 :  $r_\ell \leftarrow \alpha - c_\ell \cdot w_\ell \bmod q$ 
21 :  $\sigma = (c_1, r_1, \dots, r_k)$ 
22 : return  $(\sigma, \tilde{K}, \tilde{Z})$ 

```

Fig. 2. Signing procedure of the 2-CLSAG ring signature scheme as used in Monero

Verify($\{\text{pk}_1, \dots, \text{pk}_k\}, m, \sigma, \tilde{K}, \tilde{Z}$)
<pre> 1 : $Q \leftarrow \text{pk}_1 \parallel \dots \parallel \text{pk}_k$ 2 : $(c_1, r_1, \dots, r_k) \leftarrow \sigma$ 3 : foreach $i \in \{1, \dots, k\}$ do 4 : $(K_{ji}^o, Z_{ji}) \leftarrow \text{pk}_i$ 5 : $W_i \leftarrow \text{Hash}_{\mathbb{F}_q}(1, Q, \tilde{K}, \tilde{Z}) \cdot K_{ji}^o + \text{Hash}_{\mathbb{F}_q}(2, Q, \tilde{K}, \tilde{Z}) \cdot Z_{ji}$ 6 : endforeach 7 : $\tilde{W} \leftarrow \text{Hash}_{\mathbb{F}_q}(1, Q, \tilde{K}, \tilde{Z}) \cdot \tilde{K} + \text{Hash}_{\mathbb{F}_q}(2, Q, \tilde{K}, \tilde{Z}) \cdot \tilde{Z}$ 8 : for $i = 1, \dots, k-1$ do 9 : $c_{i+1} \leftarrow \text{Hash}_{\mathbb{F}_q}(Q, m, (r_i \cdot G + c_i \cdot W_i), (r_i \cdot \text{Hash}_{\mathbb{G}}(K_{ji}^o) + c_i \cdot \tilde{W}))$ 10 : endfor 11 : $c'_1 \leftarrow \text{Hash}_{\mathbb{F}_q}(Q, m, (r_k \cdot G + c_k \cdot W_k), (r_k \cdot \text{Hash}_{\mathbb{G}}(K_{jk}^o) + c_k \cdot \tilde{W}))$ 12 : return $c_1 = c'_1$ </pre>

Fig. 3. Verification procedure of the 2-CLSAG ring signature scheme as used in Monero

LinkAnon $_{\mathcal{A}, \Sigma}^k(\lambda)$	$\mathcal{O}_{\text{Sign}}(m, Q, \text{pk})$
<pre> 1 : $b \leftarrow_{\\$} \{0, 1\}$ 2 : foreach $i \in \{1, \dots, k\}$ do 3 : $(\text{sk}_i, \text{pk}_i) \leftarrow_{\\$} \text{KeyGen}(1^\lambda)$ 4 : endforeach 5 : $S \leftarrow \{\text{pk}_i\}_{i=1}^k$ 6 : $i_0, i_1, \text{st} \leftarrow_{\\$} \mathcal{A}(1^\lambda, \{\text{pk}_i\}_{i=1}^k)$ 7 : $b' \leftarrow_{\\$} \mathcal{A}^{\mathcal{O}_{\text{Sign}}}(\text{st})$ 8 : return $b = b'$ </pre>	<pre> 1 : if $\text{pk} \notin Q \vee Q \not\subseteq S$ then 2 : return $\perp$ 3 : endif 4 : Let i be such that $\text{pk}_i = \text{pk}$ 5 : if $\{\text{pk}_{i_0}, \text{pk}_{i_1}\} \not\subseteq Q$ $\vee \text{pk} \notin \{\text{pk}_{i_0}, \text{pk}_{i_1}\}$ then 6 : $(\sigma, \tilde{K}, \tilde{Z}) \leftarrow_{\\$} \text{Sign}(\text{sk}_i, m, Q, i)$ 7 : else 8 : Let c be such that $\text{pk}_{i_c} = \text{pk}$ 9 : $c' \leftarrow c \oplus b$ 10 : $(\sigma, \tilde{K}, \tilde{Z}) \leftarrow_{\\$} \text{Sign}(\text{sk}_{i_{c'}}, m, Q)$ 11 : endif 12 : return σ </pre>

Fig. 4. Chosen-target linkable anonymity experiment from [39]

difference between games G_i and G_j . We shall model all hash functions as random oracles and abuse notation by writing, say, ivk to interchangeably mean the label mapping an owned UTXO to the incoming viewing key and the incoming viewing key itself, and the same for other viewing keys. Finally, for clarity of exposition, we shall consider single-input single-output transactions only. The results are easily extended to transactions with multiple inputs or outputs if one wishes to restrict the spender algorithm $\mathcal{S}$ further (to account for, *e.g.*, heuristic analyses [30]).

C.1 Proof of Theorem 1

Proof (of Theorem 1). Recall from Appendix A.2 that a single-input single-output Zcash transaction is of the form

$$T = (\text{rt}, \text{cv}^{\text{net}}, C^{\text{out}}, \text{nf}, \sigma, \text{rk}, \sigma_{\text{binding}}, \pi, u'),$$

with rt , cv^{net} , C^{out} , and σ_{binding} always independent of the input note, and also σ and rk independent of the input note unless the link between rk and ak is shown in the Halo2 [16] zk-SNARK π . The proof follows by leveraging the zero-knowledge property of Halo2 to simulate π and then decouple also the tag nf (and with it u' which uses nf as a randomizer in its note commitment) from the input note. Construct the spender algorithm $\mathcal{S}$ to always set C^{out} to an encryption of a random plaintext under a random key independent of the input note [40, Section 4.20.1]. Other than that, let $\mathcal{S}$ construct T honestly, according to the Zcash protocol [11, 40], with T spending u_{i_b} only (and not $u_{i_{1-b}}$). We then have $\text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_b}) = 1$ and $\text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_{1-b}}) = 0$ independently of b and the random coins tossed by $\mathcal{S}$.

Fix $b = 0$ and consider a modification to the original game, $G_0 \mapsto G_1$, whereby, instead of running **Setup** to generate the parameters pp honestly, we generate the Halo2 zk-SNARK [16] parameters with trapdoors, such that any argument, even for a false statement, can be simulated by $\mathcal{S}$. Since Halo2 has perfect zero knowledge [16], we get $p_0 = p_1$. Let $G_1 \mapsto G_2$ be that any incoming viewing key is derived *not* from the full viewing key $\text{fvk} = (\text{ak}, \text{nk}, \text{rivk})$ as in Equation (1) but instead from an independent and uniformly random point $P_1 \leftarrow \$\mathbb{G}$ as

$$ivk \leftarrow \mathbf{x}(P_1) \bmod q, \tag{6}$$

with the proof of address integrity [40, Section 4.18.4] simulated. It is straightforward to show that, in the random oracle model, the advantage in distinguishing between ivk computed as in Equation (1) and as in Equation (6) is bounded by the probability of the distinguisher querying the random oracle $\text{Hash}_{\mathbb{G}}$ on $(\text{ak}, \text{nk}) \in \mathbb{G} \times \mathbb{F}_p$ (see [40, Section 4.2.3]), for either u_{i_0} or u_{i_1} , which is bounded by $\text{poly}(\lambda)/2^\lambda$ for a polynomial-time distinguisher. The same bound then applies to $|p_1 - p_2|$. Next, let $G_2 \mapsto G_3$ be that the dk and ovk are also sampled independently at random instead of being computed as in Equation (2). By a similar argument as in the previous step, we have $|p_2 - p_3| \leq \text{poly}(\lambda)/2^\lambda$.

Note that, at this point, $\text{ivk}(\bar{u}_{i_0})$ and $\text{ovk}(\bar{u}_{i_0})$ are independent of any other key associated with the UTXO u_{i_0} being spent in T . Write $\text{nf} \in \mathbb{F}_p$ for the tag (nullifier) of u_{i_0} as revealed in T returned by $\mathcal{S}$. It is computed (according to protocol) as in Equation (4) with $\rho, \psi \in \mathbb{F}_p$ and $\text{cm} \in \mathbb{G}$ all associated with u_{i_0} , $K \in \mathbb{G}$ fixed by the consensus protocol, and nk the *nullifier deriving key* attached to $\bar{u}_{i_0}$ [40, Section 4.16]. In particular, ρ, ψ and cm are known to the dictator in possession of the incoming viewing keys for both challenge UTXOs, *i.e.*, $\text{ivk}(\bar{u}_{i_0})$ and $\text{ivk}(\bar{u}_{i_1})$. The nullifier is, at this point, the last thing “connecting” T to the UTXO it spends. Let $\mathbb{G}_3 \mapsto \mathbb{G}_4$ be that, instead of computing the nullifier honestly, $\mathcal{S}$ sets it to $\text{nf} \leftarrow \mathbf{x}(\alpha \cdot K + \text{cm})$ for $\alpha \leftarrow \$ \{0, \dots, p-1\}$ uniformly random. $|p_3 - p_4|$ is again bounded by the probability of querying a random oracle, this time, $\text{Hash}_{\mathbb{F}_p}$ on (nk, ρ) , which itself is bounded by $\text{poly}(\lambda)/2^\lambda$. Now, for $\mathbb{G}_4 \mapsto \mathbb{G}_5$, compute the nullifier as $\text{nf} \leftarrow \mathbf{x}(P_2)$ instead, for $P_2 \leftarrow \$ \mathbb{G}$ uniformly random. Clearly, nf is now independent of u_{i_0} and $|p_4 - p_5|$ is bounded by the statistical distance Δ between a uniform distribution on $\mathbb{G}$ and a uniform distribution on $X = \{\alpha \cdot K + \text{cm} \mid \alpha \in \{0, \dots, p-1\}\} \subset \mathbb{G}$, namely,

$$\Delta = \frac{1}{2} \sum_{P \in \mathbb{G}} |\Pr[\alpha \cdot K + \text{cm} = P \mid \alpha \leftarrow \$ \{0, \dots, p-1\}] - \Pr[P_2 = P \mid P_2 \leftarrow \$ \mathbb{G}]|.$$

Recall from Appendix A.2 that for Zcash we have $q > p$ to obtain:

$$\Delta = \frac{1}{2} \sum_{P \in X} \left| \frac{1}{p} - \frac{1}{q} \right| + \frac{1}{2} \sum_{P \in \mathbb{G} \setminus X} \frac{1}{q} = \frac{p}{2} \cdot \frac{q-p}{pq} + \frac{q-p}{2} \cdot \frac{1}{q} = \frac{q-p}{q},$$

which is negligible by the Hasse bound. Specifically, we have $|q| = 2\lambda$, *i.e.*, $2^{2\lambda-1} \leq q \leq 2^{2\lambda}$, and

$$q - p \leq 2^{\lambda+1},$$

from which we conclude that

$$|p_4 - p_5| \leq \Delta = \frac{q-p}{q} \leq \frac{2^{\lambda+1}}{2^{2\lambda-1}} = 2^{-\lambda+2}.$$

At this point, everything in T is independent of u_{i_0} and the viewing keys (including the spend authorizing key used to sign T). To say that T now “spends” u_{i_1} in $\mathbb{G}_6$ is merely a conceptual change with $p_5 = p_6$. Running the above transformations in reverse, with $b = 1$, and defining games $\mathbb{G}_7, \dots, \mathbb{G}_{11}$ accordingly, we get that

$$|\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]| = |p_0 - p_{11}| \leq \frac{\text{poly}(\lambda)}{2^\lambda},$$

which is negligible. $\square$

C.2 Proof of Theorem 2

Proof (of Theorem 2). Recall from Appendix A.3 that a single-input single-output Firo transaction is of the form

$$T = (\{u_1, \dots, u_k\}, S', C', N, \pi, u'),$$

where the proofs

$$\pi = (\pi_{\text{par}}, \pi_{\text{chaum}}, \pi_{\text{bal}}, \pi_{\text{rp}})$$

will be simulated in the intermediate hybrids, which will also make the output note u' independent of the input. Also, any spender algorithm $\mathcal{S}$ that hopes to satisfy Definition 1 must clearly include both u_{i_0} and u_{i_1} in the cover set $\{u_1, \dots, u_k\}$. It will therefore suffice to reason about indistinguishability of S' , C' , and N for a transaction T spending u_{i_0} and one spending u_{i_1} . This closely follows the proof of ledger indistinguishability (LIND) in [43], except there it suffices to say that tags N are computed via a pseudorandom function of [31]. This simple argument no longer holds when the dictator gets extra information in the form of the incoming viewing keys addr_{in} .

Consider a spender algorithm $\mathcal{S}$ that produces T honestly except that, when computing the offsets S' and C' , it does not call any hash functions on s and $D = r \cdot G$ (as prescribed by [43, Section 4.8]) but samples uniformly random scalars instead. Specifically, it samples $\alpha, \beta \leftarrow \mathbb{F}_q$ and sets

$$S' \leftarrow s \cdot F - \alpha \cdot H + D \quad \text{and} \quad C' \leftarrow v \cdot G + \beta \cdot H.$$

Note that, as is the case of Zcash's C^{out} ciphertext, the network does not check if the offsets were computed according to specification or as above. We then have $\text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_b}) = 1$ and $\text{Spends}(1^\lambda, \text{pp}, L, T, \bar{u}_{i_{1-b}}) = 0$ independently of b and the random coins tossed by $\mathcal{S}$.

Fix $b = 0$ and consider a modification to the original game, $\mathbf{G}_0 \mapsto \mathbf{G}_1$, whereby, instead of running **Setup** to generate the parameters pp honestly, we generate the parameters for the proofs π with trapdoors, such that any argument, even for a false statement, can be simulated by $\mathcal{S}$.¹⁹ We have $p_0 = p_1$, as the proofs can be simulated perfectly. Let $\mathbf{G}_1 \mapsto \mathbf{G}_2$ be that S' is sampled by $\mathcal{S}$ uniformly at random from $\mathbb{G}$ instead of being computed as above. This is merely a conceptual change, as S' is uniformly random and independent of D in either case. We therefore have $p_1 = p_2$. Let $\mathbf{G}_2 \mapsto \mathbf{G}_3$ be that the tag N is chosen uniformly at random from $\mathbb{G}$ instead of being computed as in Equation (5). By the Decisional Diffie-Hellman Inverse (DDHI) assumption [19, 31], there exists a negligible function $\varepsilon_{\text{ddhi}}$ such that

$$|p_2 - p_3| \leq n \cdot \varepsilon_{\text{ddhi}}. \quad (7)$$

Indeed, otherwise, given a DDHI triple $(B, A = a \cdot B, \tilde{A}) \in \mathbb{G}^3$, one could simulate $\mathcal{D}$'s view as follows and use it to decide if $\tilde{A} = (1/a) \cdot B$ or if it was sampled independently from $\mathbb{G}$. Sample $s_1, r \leftarrow \mathbb{F}_q$, parts of the spending key addr_{sk} associated to u_{i_0} , honestly and set $D \leftarrow r \cdot G$. Sample $\gamma \leftarrow \mathbb{F}_q$ and produce the generators K and F as

$$K \leftarrow B + D \quad \text{and} \quad F \leftarrow \gamma \cdot B.$$

¹⁹ Since the balance proof π_{bal} is just a Schnorr-like proof of knowledge of multiple discrete logarithms to the same base, to simulate it, one may instead program the random oracle responses.

Set the tag of T returned by $\mathcal{S}$ to $N \leftarrow \tilde{A}$ and the serial number commitment in u_{i_0} to

$$S \leftarrow \gamma \cdot A + D.$$

Sample uniform $w \leftarrow_{\$} \mathbb{F}_q$ and program the random oracles to have

$$\text{Hash}_{\mathbb{F}_q}(3, z) + \text{Hash}_{\mathbb{F}_q}(2, s_1, i) = w,$$

for z the random nonce in u_{i_0} and set $P_2 \leftarrow S - w \cdot F$. To then simulate any other UTXO $\hat{u}$ under the control of addr_{sk} , sample the nonce $\hat{z}$ honestly, sample $\hat{w} \leftarrow_{\$} \mathbb{F}_q$, and program the random oracles to have

$$\text{Hash}_{\mathbb{F}_q}(3, \hat{z}) + \text{Hash}_{\mathbb{F}_q}(2, s_1, i) = \hat{w}.$$

Finally, set the serial number commitment to

$$\hat{S} \leftarrow \hat{w} \cdot F + P_2.$$

If $\tilde{A} = (1/a) \cdot B$, then the view of $\mathcal{D}$ is consistent with G_2 ; otherwise, it is consistent with G_3 , so $\mathcal{D}$ telling apart the two games can be leveraged to break the DDHI assumption. The factor of n in Equation (7) comes from the fact that our reduction must guess ahead of time the index i_0 that $\mathcal{D}$ is going to return.

At this point, everything in T is independent of u_{i_0} and the associated viewing key $\text{addr}_{\text{in}}(u_{i_0})$. To say that T now “spends” u_{i_1} in G_4 is another conceptual change with $p_3 = p_4$. Running the above transformations in reverse, with $b = 1$, and defining games $G_5, \dots, G_7$ accordingly, we get that

$$|\Pr[b' = 1 \mid b = 0] - \Pr[b' = 1 \mid b = 1]| = |p_0 - p_7| \leq 2 \cdot n \cdot \varepsilon_{\text{ddhi}},$$

which is negligible. $\square$